Smart KYC

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In

Computer Engineering

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April 2025

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Smart KYC

is the bonafide work of

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Carried out in the partial fulfillment of the degree of Bachelor of Technology in Computer Engineering at Dharmsinh Desai University in the academic session

December 2024 to April 2025

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TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Manthankumar Patel is** working with **DRC Systems India Ltd** at **Gandhinagar** as an **Intern** from **1st January,2025** till date. During this period, he has successfully completed the **"Smart KYC"** project under the mentorship of **Mr. Parth Kher**.

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Smart KYC

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1. Introduction

1.1 Abstract

Smart KYC is an AI-driven Know Your Customer (KYC) verification system designed to streamline and automate the identity verification process using modern technologies. Built with FastAPI for the backend and React for the frontend, the system processes official identity documents such as Aadhaar and PAN cards using Optical Character Recognition (OCR) powered by OpenAI Vision. It extracts and verifies key user details including name, date of birth, Aadhaar number, PAN card number, gender, and address.

To ensure authenticity, Smart KYC implements facial recognition through DeepFace, matching the user's selfie with the photo present on submitted documents.

Additional features include a robust profile management module, OTP-based verification for email and phone, secure image upload and deletion, and comprehensive form validations. By combining AI, facial verification, and intelligent document processing, Smart KYC enhances security, improves user experience, and significantly reduces manual intervention in identity verification processes.

1.2 Introduction of Smart KYC

Our KYC Verification System is an advanced identity verification solution that automates the process of verifying users through document processing and facial recognition. It extracts key details from official identity documents such as Aadhaar and PAN, ensuring accuracy and compliance with regulatory standards. By eliminating manual verification, the system makes KYC faster, more reliable, and highly secure.

This system is useful because it reduces errors, prevents fraud, and enhances security during the identity verification process. With features like JWT-based authentication, OTP verification, and biometric face matching, it ensures that only genuine users can access services. Automating KYC saves time and resources for both users and businesses while improving compliance with legal requirements.

The KYC Verification System is ideal for banks, financial institutions, fintech companies, and businesses that require secure and efficient identity verification. It streamlines customer onboarding, fraud prevention, and regulatory compliance, making it an essential tool for organizations handling sensitive user data.

1.3 Problem Definition

Identity verification is a crucial process for businesses, financial institutions, and government services to prevent fraud and ensure regulatory compliance. However, traditional KYC (Know Your Customer) methods are manual, time-consuming, and error-prone, leading to delays in customer onboarding and increased operational costs. Users often need to submit physical documents, which organizations must verify manually, making the process slow and inefficient.

Challenges in Traditional KYC Systems

- **Time-Consuming & Inefficient** – Manual verification of documents leads to long processing times.
- **Fraud & Security Risks** – Fake or tampered documents can be used for identity fraud, and weak authentication systems increase vulnerability.
- **High Operational Costs** – Organizations spend significant resources on manual verification and compliance.
- **Regulatory Compliance Issues** – Strict KYC regulations require careful verification, which can be complex and resource-intensive.

1.4 Proposed Solution

To address the inefficiencies and security risks in traditional KYC processes, our automated KYC Verification System leverages advanced technologies such as OCR, facial recognition, and secure authentication methods. This system streamlines identity verification, reduces manual efforts, and enhances security, making the KYC process faster, more accurate, and cost-effective.

- **Automated Document Processing** – Extracts and verifies details from Aadhaar, PAN, and other documents using OCR technology, eliminating manual entry errors.
- **Facial Recognition for Identity Matching** – Ensures the authenticity of users by comparing their selfie with document photos, preventing identity fraud.
- **Secure Authentication with JWT & OTP** – Implements token-based authentication and OTP verification to enhance security and prevent unauthorized access.
- **Faster & Efficient Customer Onboarding** – Reduces processing time by automating verification, allowing businesses to onboard customers quickly.

2. About System

2.1 Purpose

Our KYC Verification System is designed to simplify and secure the identity verification process for businesses, financial institutions, and regulatory bodies. Traditional KYC methods are slow, error-prone, and costly. This system automates document processing and facial recognition to ensure quick and accurate verification.

Key objectives include:

- **Automated Document Verification** – Extracts and validates Aadhaar, PAN, and other documents.
- **Seamless User Experience** – Provides a smooth and hassle-free verification process.
- **Faster Onboarding & Cost Efficiency** – Reduces processing time and operational costs.

2.2 Objectives

The primary objective of our KYC Verification System is to provide a fast, secure, and automated identity verification solution. By leveraging AI-driven document processing and facial recognition, the system ensures efficient and accurate verification while reducing manual effort.

Key objectives include:

- **Automate KYC Process** – Minimize manual intervention by automating document extraction and verification.
- **Enhance Security** – Implement robust authentication measures to prevent unauthorized access.
- **Improve Efficiency** – Reduce processing time and streamline customer onboarding.
- **Ensure Data Privacy** – Protect sensitive user information with secure data handling.
- **Scalability & Adaptability** – Support various industries and compliance requirements.

2.3 Scope

The KYC Verification System is designed to streamline and secure identity verification processes across various industries. It automates document verification, facial recognition, and authentication to ensure accuracy and compliance.

Key Areas of Scope:

- **User Registration & Authentication** – Secure onboarding using OTP and JWT-based authentication.
- **Document Verification** – Extracts and validates details from Aadhaar, PAN, and other identity documents.
- **Facial Recognition** – Matches user selfies with official documents for enhanced security.
- **Compliance Management** – Ensures adherence to KYC regulations for financial institutions and businesses.
- **Scalability** – Can be integrated into various platforms, including banking, insurance, and fintech applications.

This system provides a fast, secure, and automated approach to identity verification, reducing fraud and manual effort.

2.4 Tools and Technologies

The KYC Verification System is built using a combination of modern tools and technologies to ensure secure, efficient, and automated identity verification. These technologies are categorized based on their specific role in the system.

Frontend Technologies

The frontend provides an interactive user interface for document submission, verification status, and user authentication.

- **ReactJS** – A JavaScript library used for creating a dynamic, responsive, and user-friendly interface. Improve user experience with real-time updates.

Backend Technologies

The backend is responsible for processing user requests, authentication, and managing the KYC workflow.

- **FastAPI** – A high-performance web framework for building APIs quickly and efficiently, ensuring fast request handling.
- **MIMEText** – A Python library used for composing and sending email notifications, including OTPs and verification confirmations.

Database Technologies

The system requires a structured and secure database to store user details, documents, and verification logs.

- **SQLAlchemy** – A powerful ORM (Object Relational Mapper) that simplifies database interactions and ensures data consistency.
- **Alembic** – A database migration tool for handling schema changes efficiently.

OCR & AI-Based Processing

To automate document verification, Optical Character Recognition (OCR) and AI-powered tools are used for text extraction and face matching.

- **OpenAI** – Used for AI-based text processing and document analysis.
- **PaddleOCR** – A robust OCR tool optimized for extracting text from Aadhaar, PAN, and other identity documents.
- **EasyOCR** – A lightweight OCR tool that improves text recognition accuracy in scanned documents.
- **TesseractOCR** – An open-source OCR engine used for extracting text from various document formats.
- **DeepFace** – A deep-learning-based facial recognition library for verifying users by matching their selfie with ID documents.

3. Analysis

The analysis phase is crucial in understanding the requirements, functionalities, and workflow of the KYC Verification System. This phase includes identifying the system's objectives, user interactions, data flow, and security measures to ensure an efficient and secure verification process.

3.1 Functional Analysis

The system consists of various functionalities designed to enhance the verification process:

- **User Registration & Authentication** – Users register and log in securely using JWT-based authentication and OTP verification.
- **Document Upload & Processing** – Users upload identity documents, which the system processes using OCR and AI.
- **Data Extraction & Verification** – The system extracts key details (name, DOB, Aadhaar/PAN number, etc.) and verifies them against user input.
- **Face Matching** – DeepFace technology compares the user's selfie with the photo on their identity document to ensure authenticity.
- **Verification Status & Notifications** – Users receive real-time updates on their verification status through email notifications.

3.2 Security & Compliance Analysis

The system prioritizes security to protect sensitive user information:

- **JWT Authentication** – Ensures secure access control.
- **End-to-End Encryption** – Protects data during transmission.
- **Regulatory Compliance** – Adheres to KYC guidelines to prevent fraud and identity theft.

3.3 Functional Requirements

1. **Users:**

   R1.1: Register

   Descriptions:

   Users can create a new account by providing their email, password, and other necessary details. The system will validate the information and register the user.

Input:

Email, password, name, phone number, gender and other required field

Output:

Successful account creation message, redirect to login page.

R1.2: Login

Description:

Users can log in using their credentials (email and password). Upon successful login, they are redirected to the dashboard.

Input:

Email and password.

Output:

Redirect to user dashboard upon successful login, error message for incorrect credentials.

R1.3: Forget Password

Description:

Users can reset their password by entering their registered email. The system will send a reset password link via email.

Input:

Nothing. (Just click on forget password button)

Output:

Email with reset password link sent to your registered email address.

R1.4: Change Password

Description:

After login, users can change their password. They will need to provide their old password and enter a new one.

Input:

Current password, new password, and confirm new password.

Output:

Successful password change confirmation.

R1.5: Edit Profile

Description:

Users can update their profile details such as name, email (if allowed), phone number, etc., after logging in.

Input:

New profile details (name, email, phone number, etc.).

Output:

Successful profile update confirmation.

R1.6: Upload or Edit Profile Photo

Description:

Users can upload a new profile photo or update their existing photo. The photo will be saved on the server.

Input:

Image file (JPG, JPEG, PNG, GIF).

Output:

Successful profile Photo update/upload confirmation.

R1.7: Delete Profile Photo

Description:

Users can delete their profile photo. If a photo exists, it will be removed from the server and database.

Input:

No input (request to delete the current profile photo).

Output:

Confirmation message that the profile photo has been deleted.

R1.8: Verify Email Using OTP Verification

Description:

Upon registration or when requested, the system will send an OTP to the user's email to verify their identity. The user must enter the OTP to complete the verification.

Input:

Just click on the verify email button and write OTP which is received on email.

Output:

The system will display a success message if the OTP is correct, or a failure message if the OTP is incorrect

R1.9: KYC Verification via Document Upload

Description:

Users need to upload identity proof documents (e.g., Aadhaar, PAN) as part of the KYC process. The system will verify the documents before proceeding with verification.

Input:

Document file (Aadhaar, PAN, etc.).

Output:

Confirm successful document upload and verification status.

R1.10: KYC Face Verification

Description:

Users upload a selfie or take one, and the system matches the face in the uploaded photo with the face in the identity documents (Aadhaar or PAN). If the faces match, the KYC is completed.

Input:

Selfie image file, document images (Aadhaar or PAN).

Output:

Face verification result, successful or failed match message.

R1.11: Update KYC Status

Description:

After verifying both the documents and face, the system updates the KYC status of the user (Pending, Verified, Rejected).

Output:

Updated KYC status (Pending, Verified, Rejected).

3.4 Non-Functional Requirements

Performance:

The system should provide quick response times for actions like login, registration, and profile updates, ideally under 2 seconds. It should scale to handle at least 1000 concurrent users using load balancing and distributed databases. File uploads and system latency should remain optimal even under heavy load.

Scalability:

The system must be capable of growing as user numbers and data volumes increase, utilizing horizontal and vertical scaling. Cloud solutions, modular feature design, and efficient resource management are key to maintaining performance as the platform expands.

Security:

Strong encryption must be used for data at rest (bcrypt/Argon2 for passwords) and in transit (TLS/SSL for communication). two-factor authentication (2FA), and regular security audits ensure data protection. Secure logging will monitor suspicious activity.

Availability:

The system should maintain 99.9% uptime, minimizing downtime to less than 8 hours per year. Redundancy, failover techniques, and replication of critical components (databases, servers) will ensure high availability. Regular maintenance and monitoring will keep the system operational.

Usability:

The system should offer an intuitive, user-friendly interface with minimal steps to complete tasks. It should be fully responsive and accessible across devices, ensuring a seamless experience on both desktop and mobile. Regular usability testing will help improve the user experience.

3.5 Hardware Requirements

Server Specifications:

- **CPU**: Multi-core processor (minimum 4 cores, recommended 8 or more cores for high traffic systems).
- **RAM**: Minimum **16 GB RAM** for handling simultaneous user requests.
- **Storage**: At least **1 TB SSD storage** for fast data access and scalability.

Database Server:

- **CPU**: Multi-core processor (4 cores or more).
- **RAM**: Minimum **16 GB RAM** for optimal query processing.
- **Storage**: **1 TB SSD** with database replication for high availability.

Web Server:

- **CPU**: Multi-core processor (4 cores or more for handling web requests).
- **RAM**: Minimum **8-16 GB RAM** to support the web application.
- **Storage**: **500 GB SSD** for website assets and logs.

3.6 Class Diagram

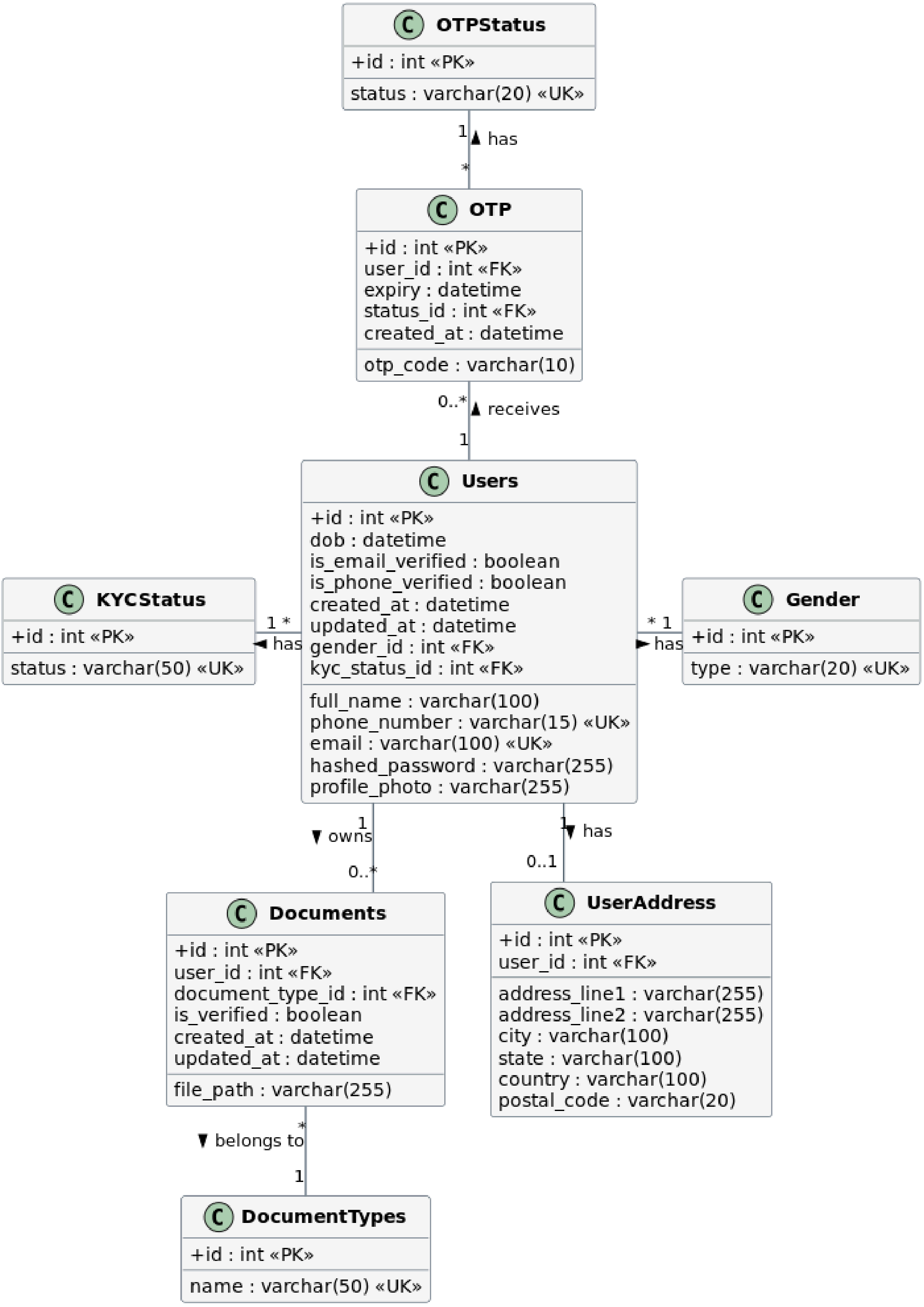

Fig 3.6.1 Class Diagram

3.7 Sequence Diagram

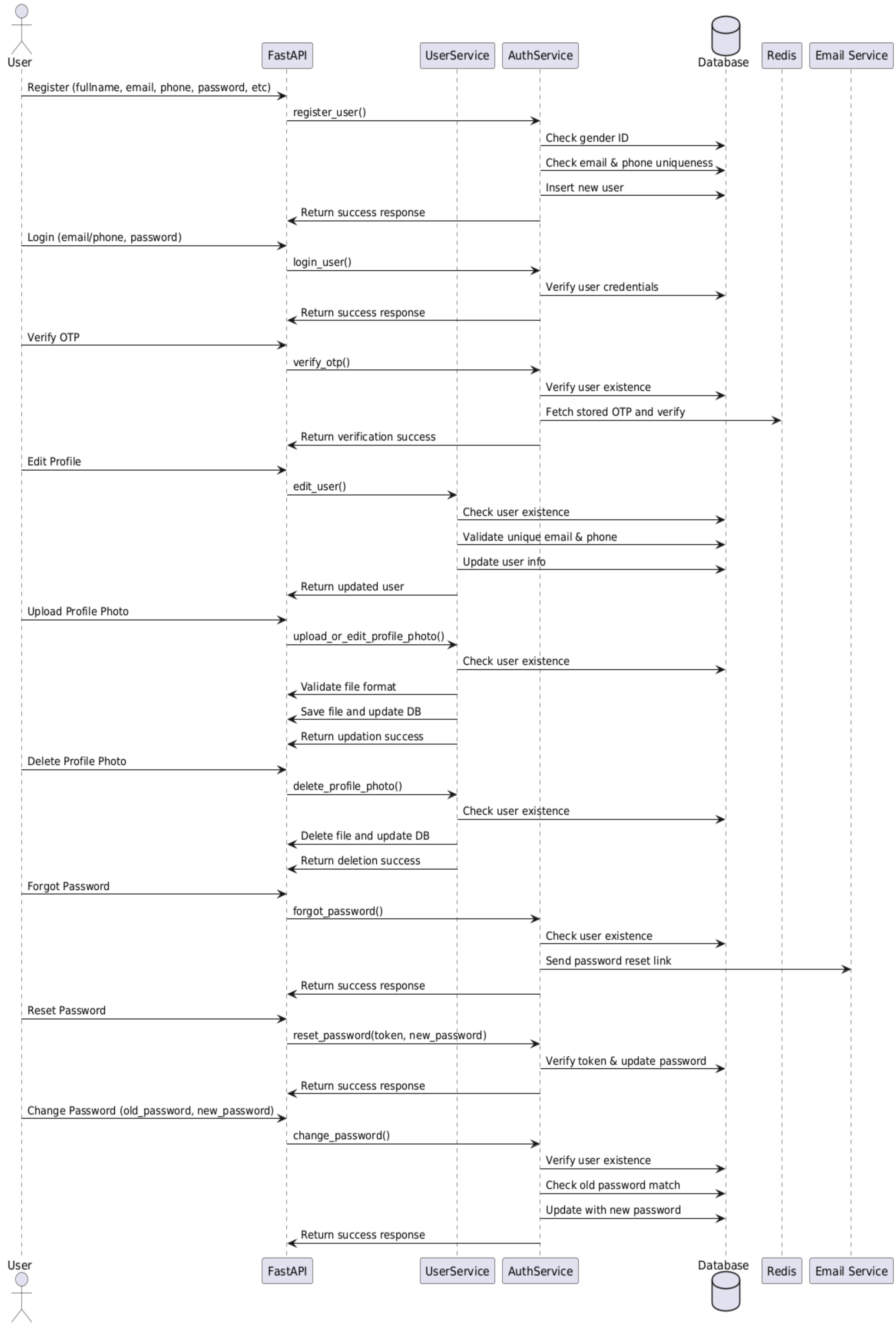

Fig 3.7.1 Authentication Sequence Diagram

3.8 Activity Diagram

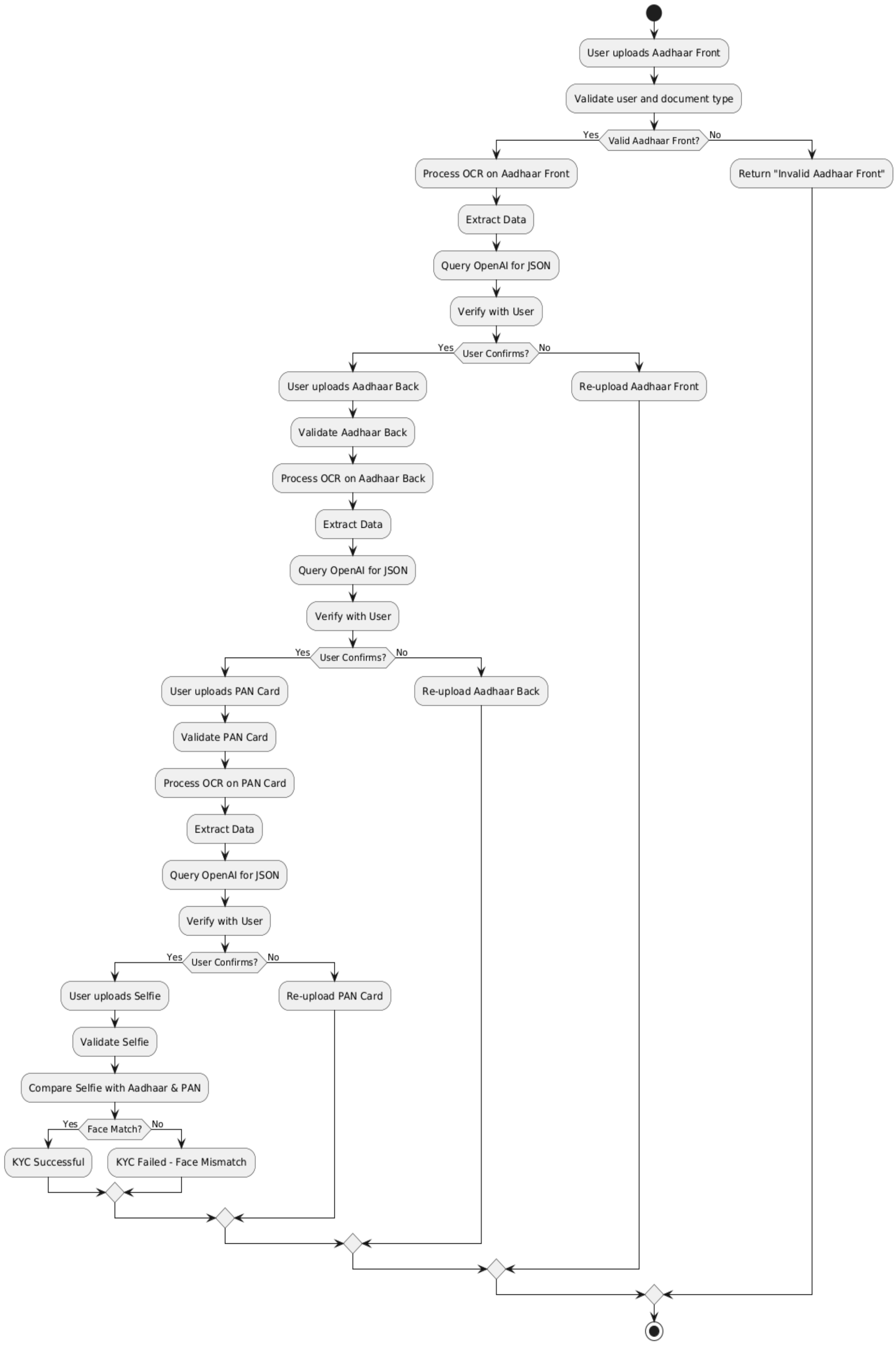

Fig 3.8.1 KYC Process Activity Diagram

3.9 Use Case Diagram

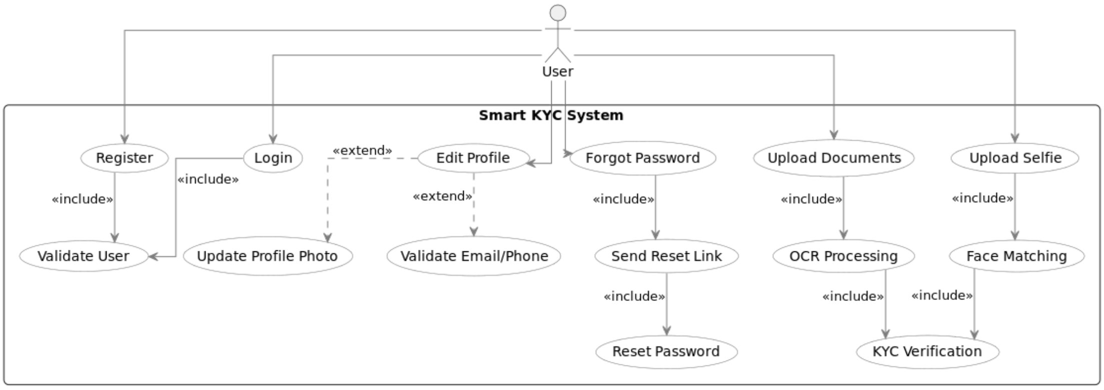

Fig 3.9.1 Use Case Diagram

3.10 Deployment Diagram

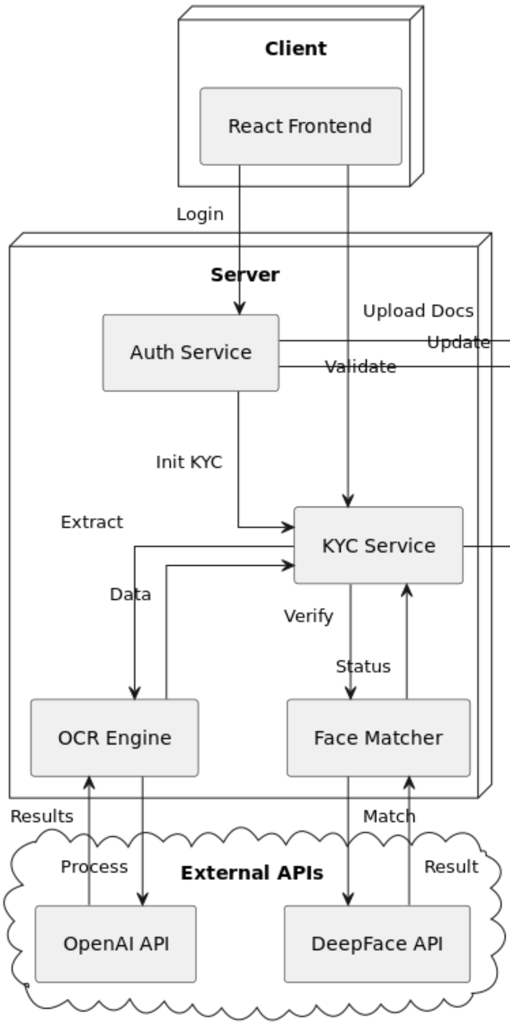

Fig 3.10.1 Deployment Diagram

4. Design

4.1 Database (Data Dictionary)

- alembic_version

Field Name	Data Type	Constraints	Description
version_num	VARCHAR(32)	PRIMARY KEY	Alembic migration version identifier

Table 4.1.1 Alembic Version Table

- document_details

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for document details
document_id	INT	FOREIGN KEY (documents.id), NOT NULL	Reference to documents table
details	JSON	NULL	Extracted document details in JSON format
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
updated_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record last update timestamp

Table 4.1.2 Document Details Table

- document_type

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for document type
name	VARCHAR(50)	UNIQUE, NOT NULL	Document type name (Aadhaar Front, PAN, Selfie, etc.)

Table 4.1.3 Document Type Table

- documents

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for document
user_id	INT	FOREIGN KEY (users.id), NOT NULL	Reference to the user who uploaded the document
document_type_id	INT	FOREIGN KEY (document_type.id), NOT NULL	Reference to document type
file_path	VARCHAR(255)	NOT NULL	File storage path
is_verified_document	TINYINT(1)	NULL	Document verification status (1 = verified, 0 = not verified)
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
updated_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record last update timestamp

Table 4.1.4 Document Table

- gender_type

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for gender type
type	VARCHAR(20)	UNIQUE, NOT NULL	Gender type (Male, Female, Other)

Table 4.1.5 Gender Type Table

- kyc_status

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for KYC status
status	VARCHAR(50)	UNIQUE, NOT NULL	KYC status (Pending, Verified, Rejected)

Table 4.1.6 KYC Status Table

- otp

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for OTP entry
user_id	INT	FOREIGN KEY (users.id), NOT NULL	Reference to user who received OTP
otp	VARCHAR(10)	NOT NULL	One-time password value
expiry	DATETIME	NOT NULL	OTP expiry time
otp_status_id	INT	FOREIGN KEY (otp_status.id), NOT NULL	Reference to OTP status
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Table 4.1.7 OTP table

- otp_status

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for OTP status
status	VARCHAR(20)	UNIQUE, NOT NULL	OTP status (Sent, Not Sent, Verified)

Table 4.1.8 OTP Status Table

- user_address

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for user address
user_id	INT	FOREIGN KEY (users.id), NOT NULL	Reference to user
address_line1	VARCHAR(255)	NOT NULL	First line of address
address_line2	VARCHAR(255)	NULL	Second line of address
city	VARCHAR(100)	NOT NULL	City name
state	VARCHAR(100)	NOT NULL	State name
country	VARCHAR(100)	NOT NULL	Country name
postal_code	VARCHAR(20)	NOT NULL	Postal code / ZIP code

Table 4.1.9 User Address Table

- users

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for user
full_name	VARCHAR(100)	NOT NULL	User's full name
phone_number	VARCHAR(15)	UNIQUE, NOT NULL	User's phone number
email	VARCHAR(100)	UNIQUE, NOT NULL	User's email address
dob	DATETIME	NOT NULL	User's date of birth

hashed_password	VARCHAR(255)	NOT NULL	Hashed password for authentication
profile_photo	VARCHAR(255)	NULL	User's profile photo file path
is_email_verified	TINYINT(1)	NOT NULL	Email verification status (1 = verified, 0 = not verified)
is_phone_verified	TINYINT(1)	NOT NULL	Phone verification status (1 = verified, 0 = not verified)
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
updated_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record last update timestamp
gender_id	INT	FOREIGN KEY (gender_type.id), NOT NULL	Reference to gender type
kyc_status_id	INT	FOREIGN KEY (kyc_status.id), NULL	Reference to KYC status

Table 4.1.10 Users Table

4.2 Frontend Interface

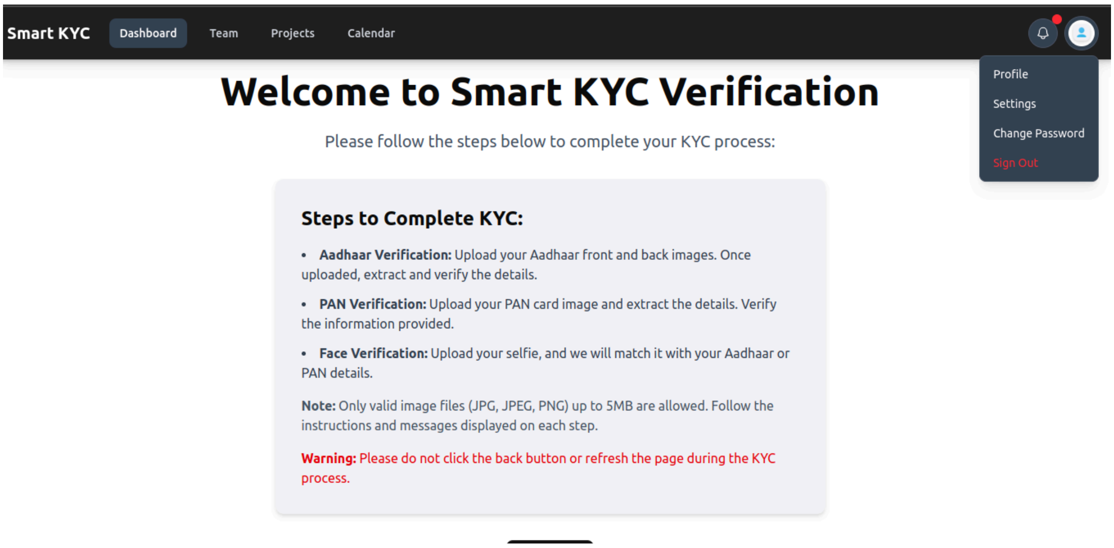

Fig 4.2.1 Welcome page from Smart KYC verification

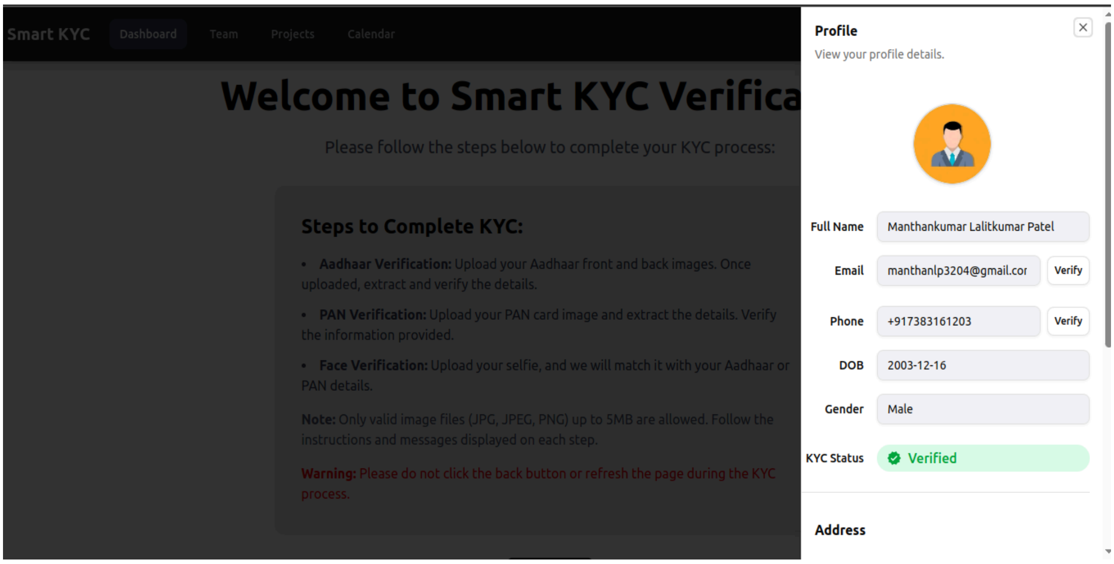

Fig 4.2.2 Profile Page

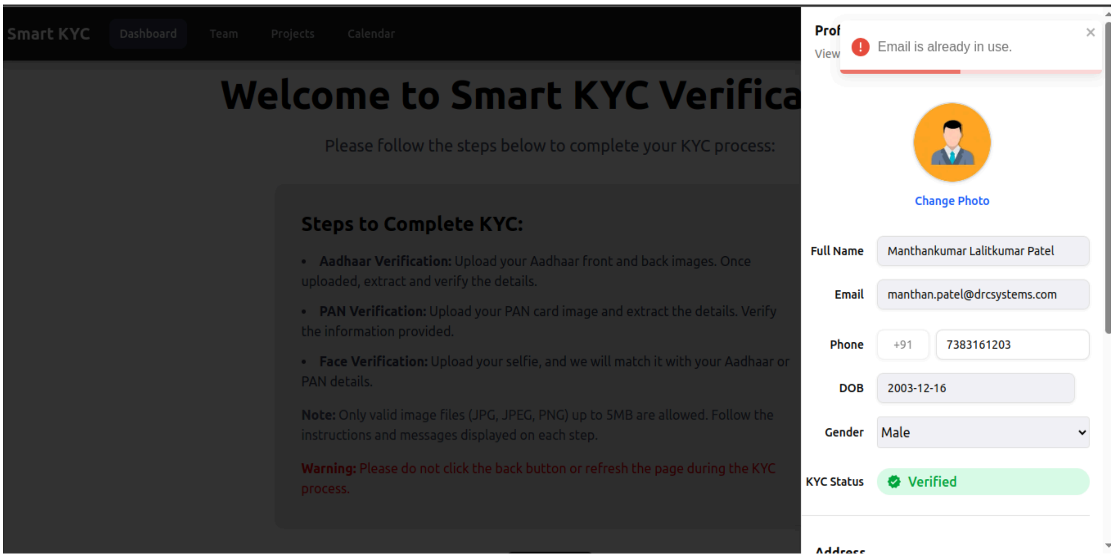

Fig 4.2.3 Profile update failed: Email exists.

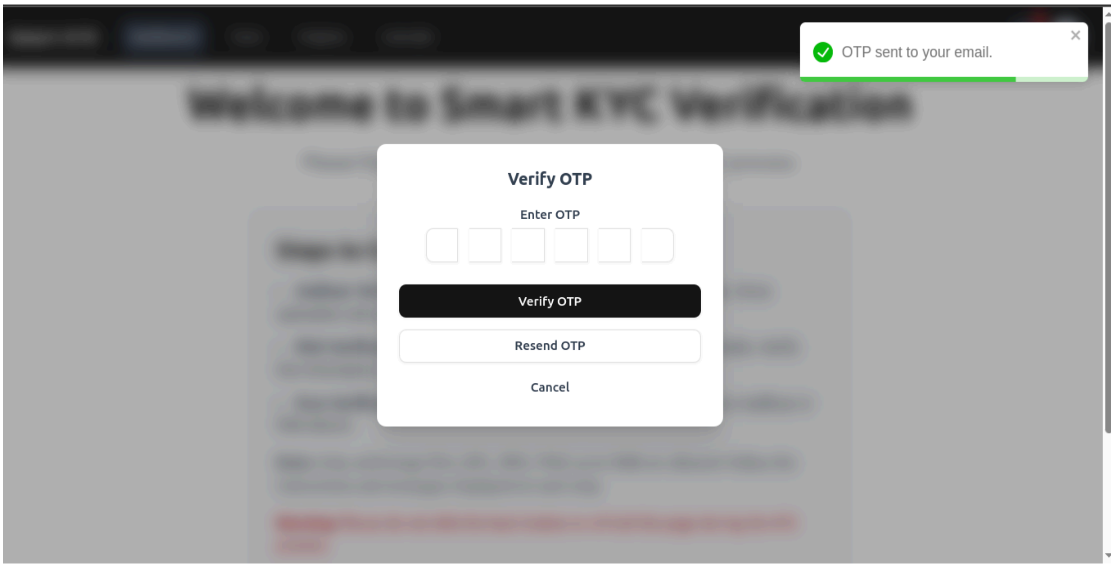

Fig 4.2.4 Enter OTP to verify email.

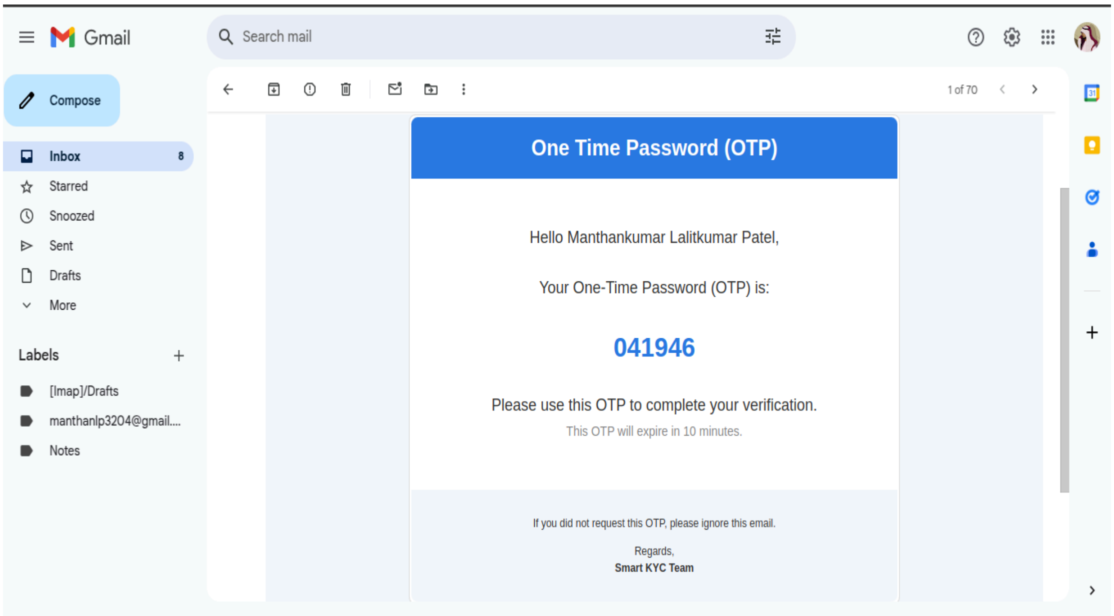

Fig 4.2.5 OTP sent to registered email.

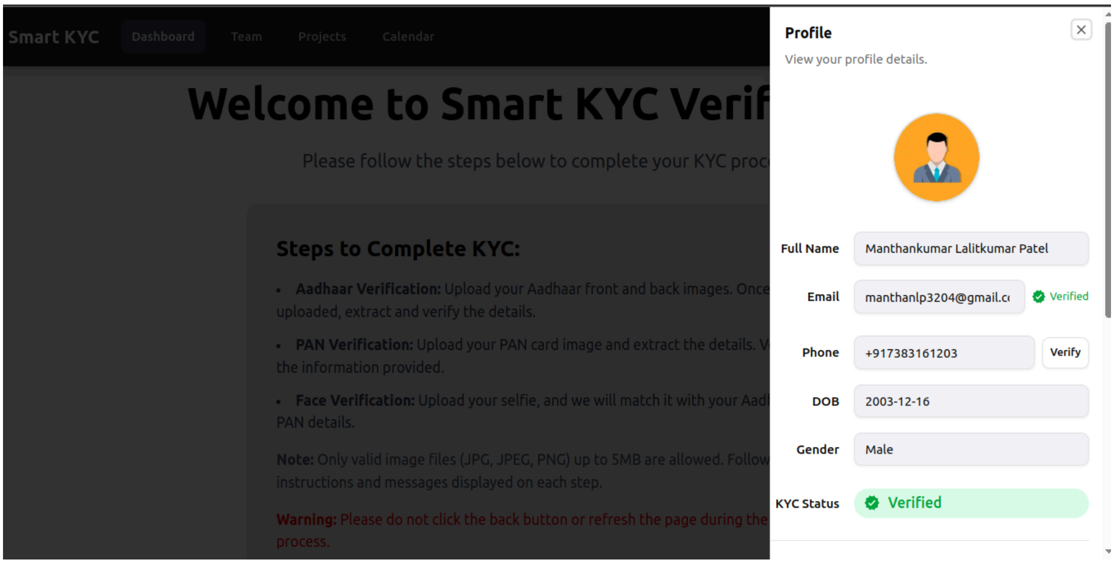

Fig 4.2.6 After email verify successfully using OTP

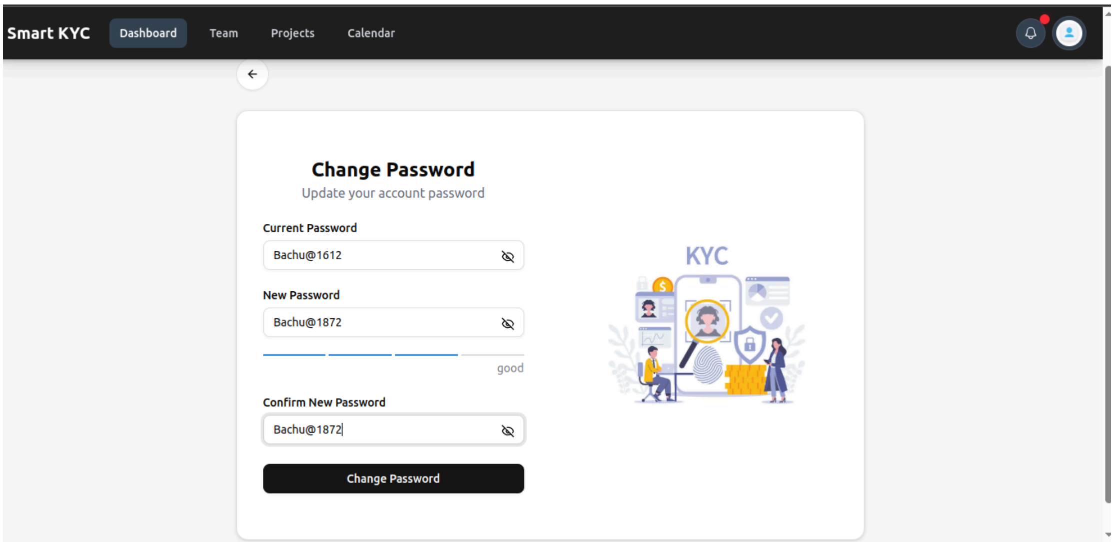

Fig 4.2.7 Change password Page

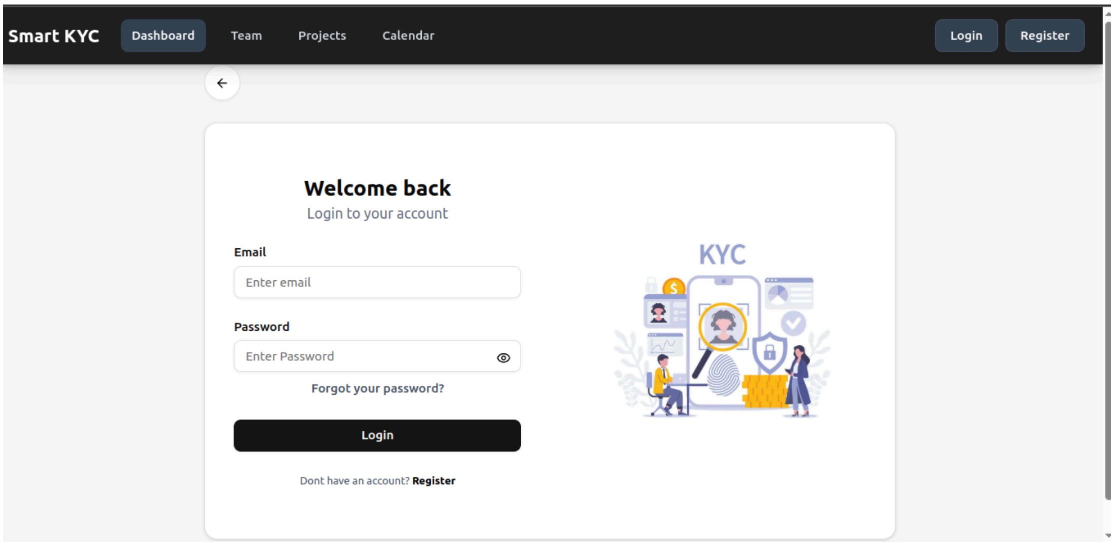

Fig 4.2.8 Login Page

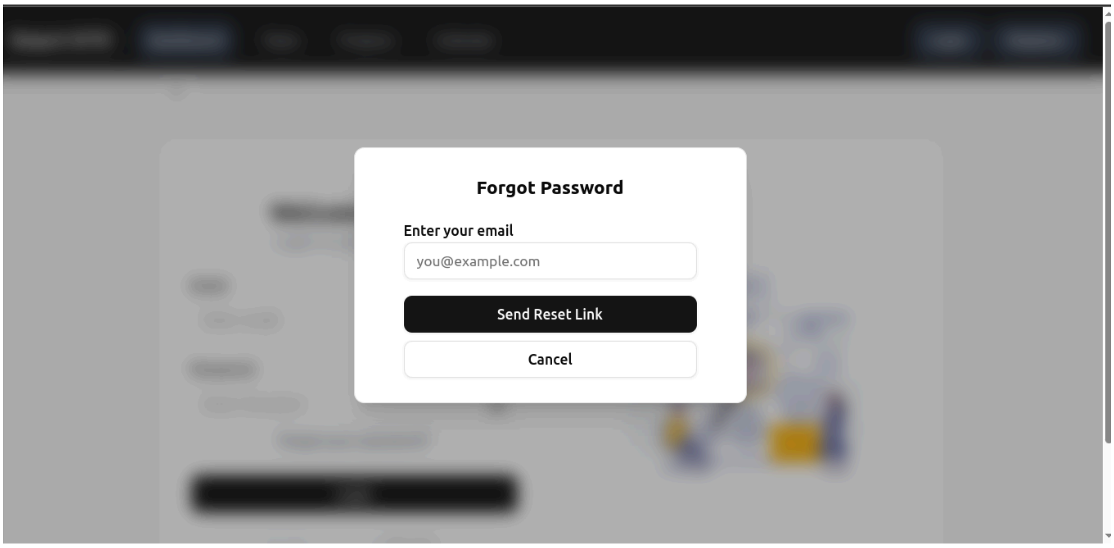

Fig 4.2.9 Forgot password Page

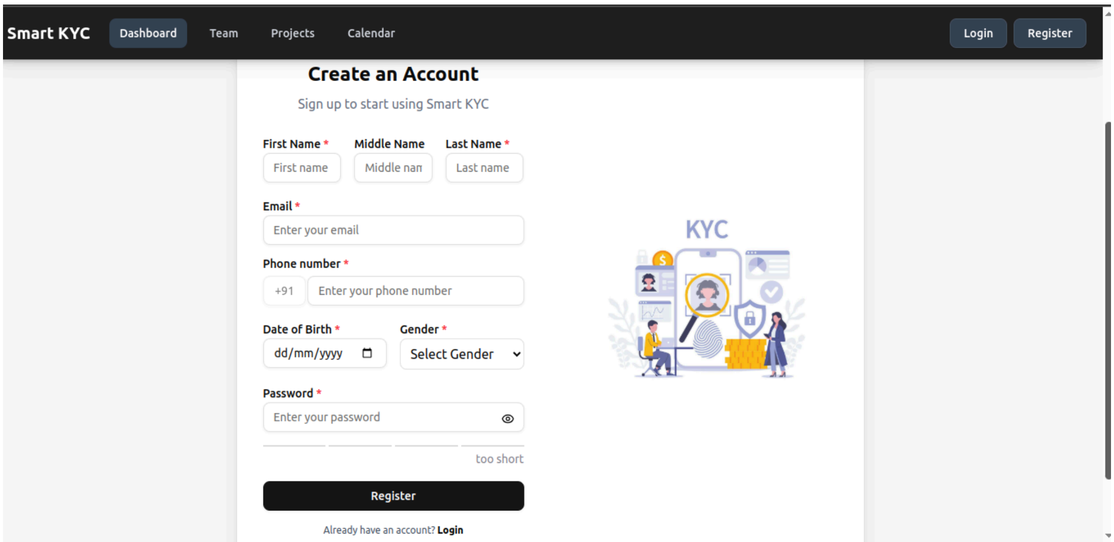

Fig 4.2.10 Register Page

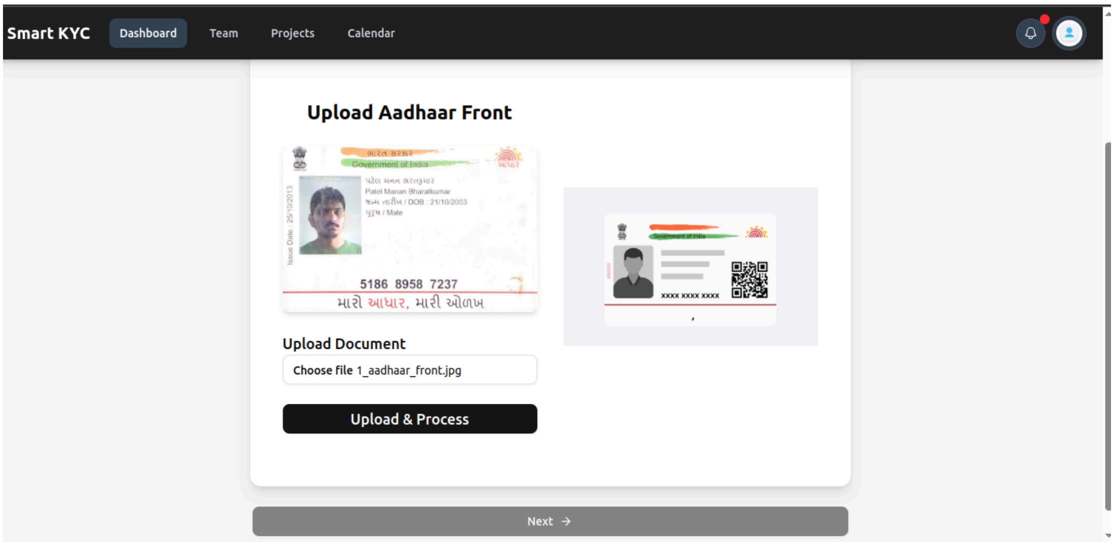

Fig 4.2.11 Aadhaar card front upload and process Page

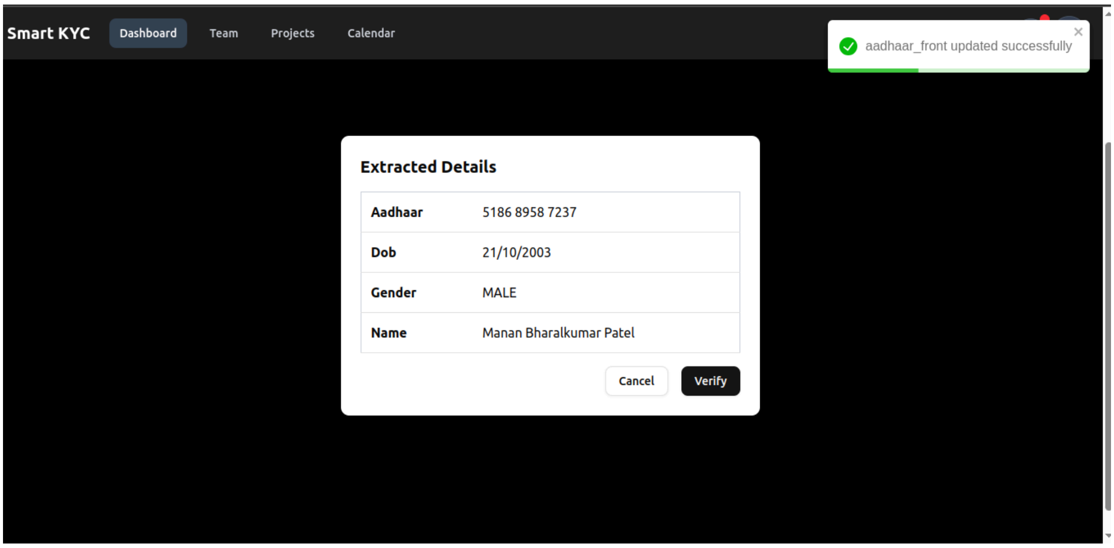

Fig 4.2.12 Aadhaar card front detail verification Page

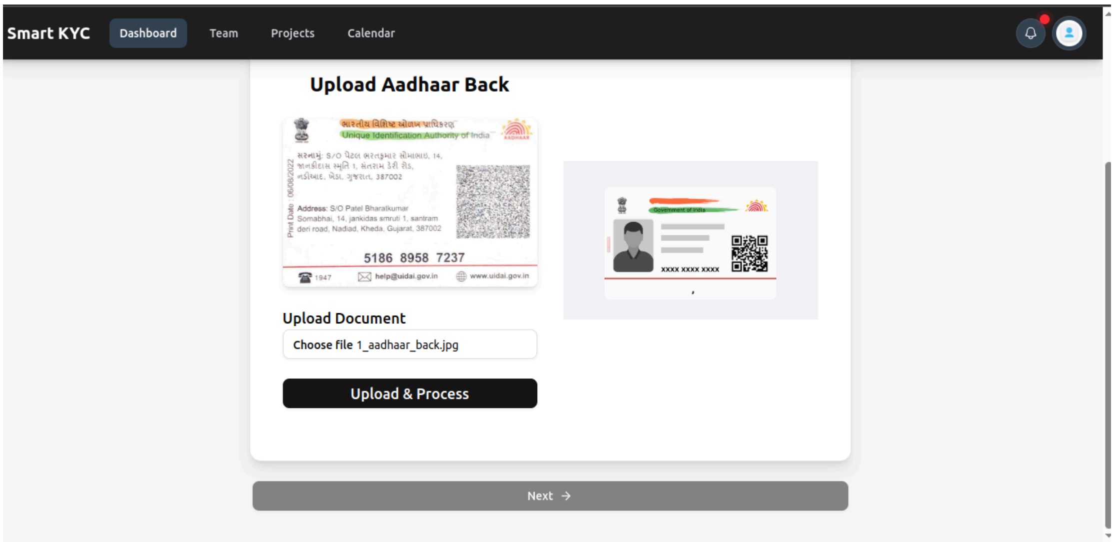

Fig 4.2.13 Aadhaar card back upload and process Page

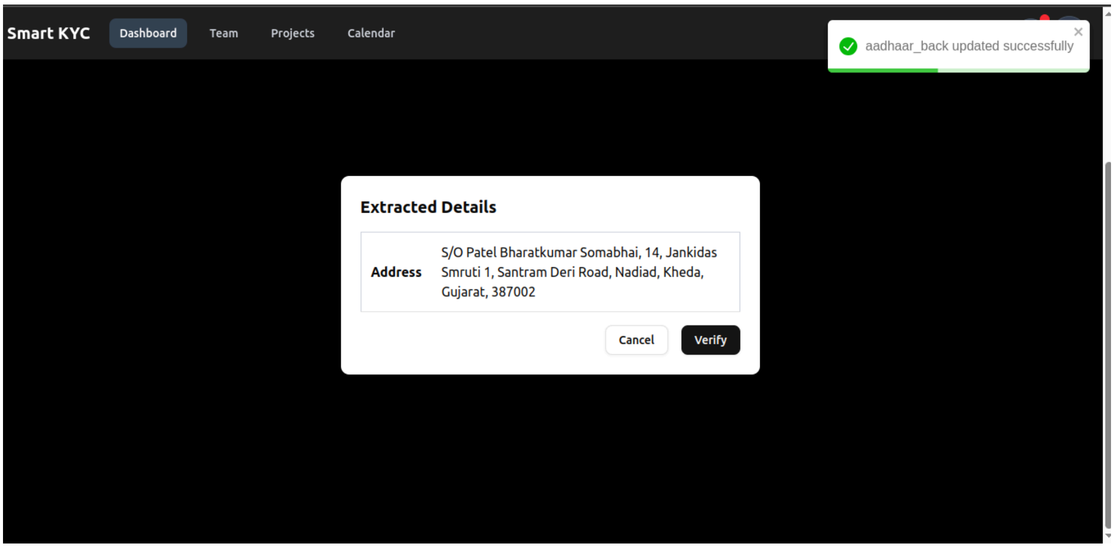

Fig 4.2.14 Aadhaar card back detail verification Page

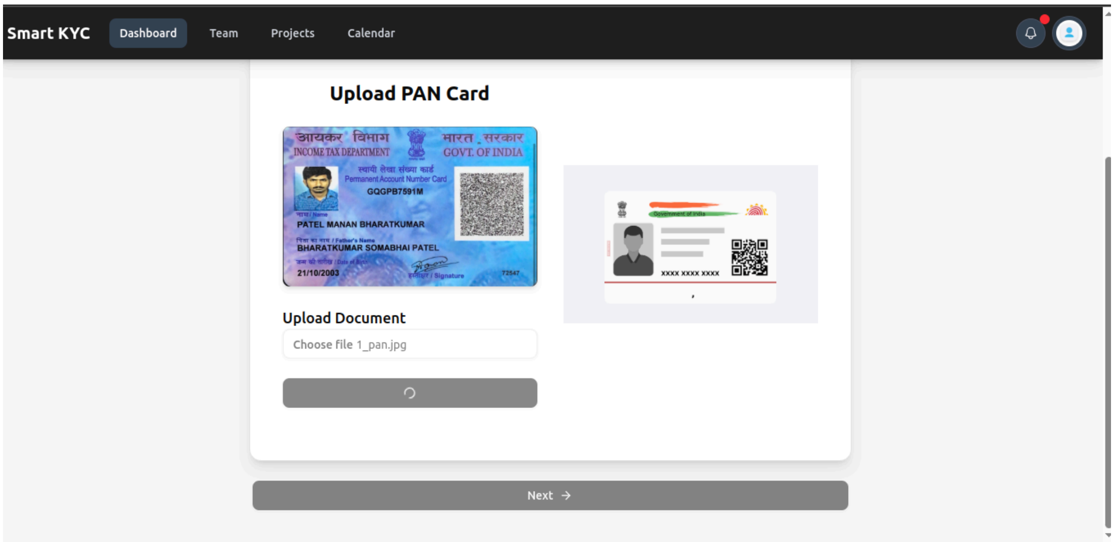

Fig 4.2.15 Pan card upload and process Page

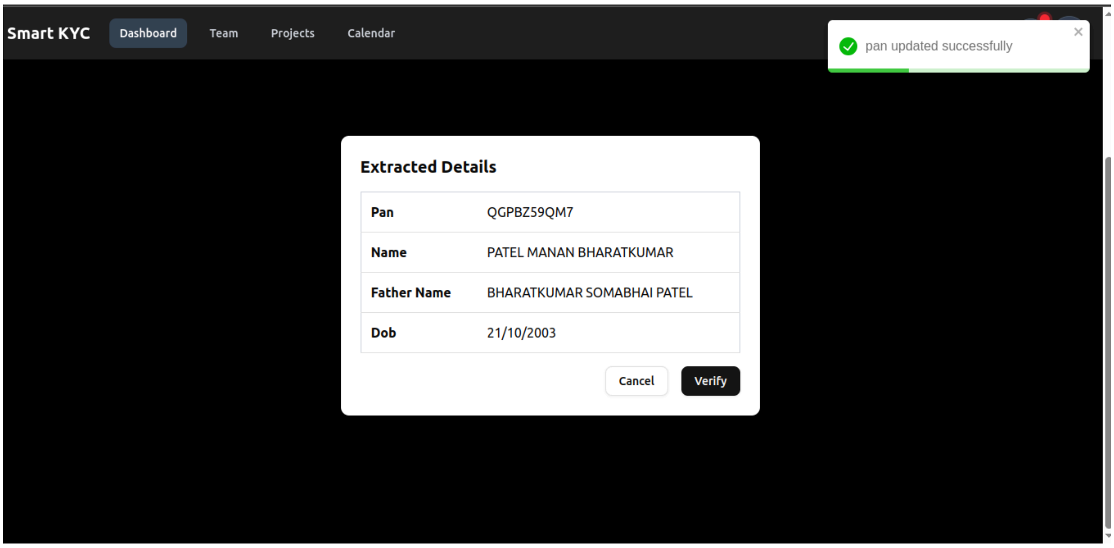

Fig 4.2.16 Pan card detail verification Page

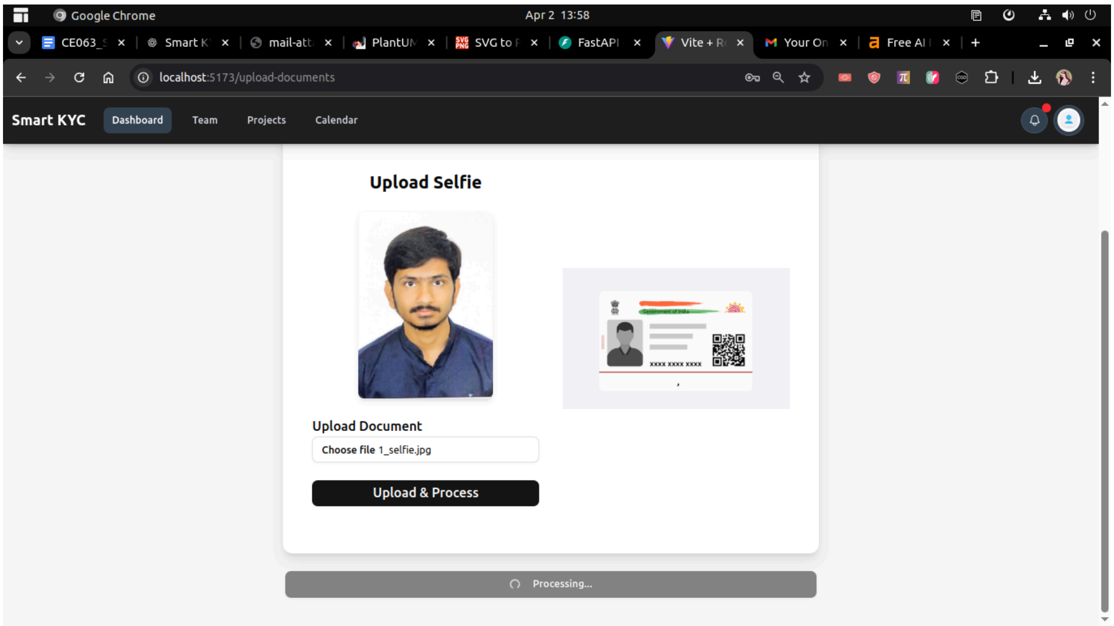

Fig 4.2.17 Upload selfie Page

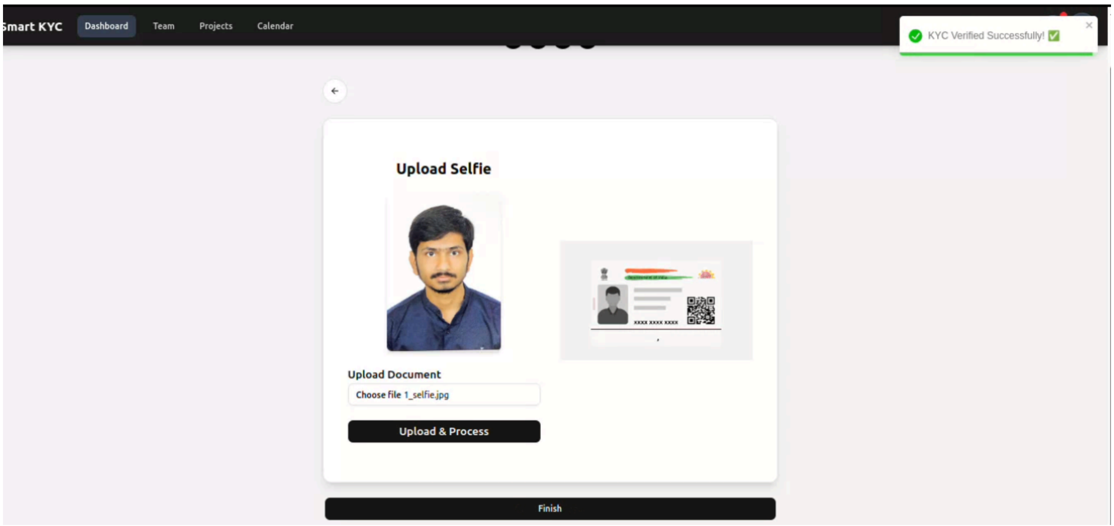

Fig 4.2.18 Selfie matched with ID proof

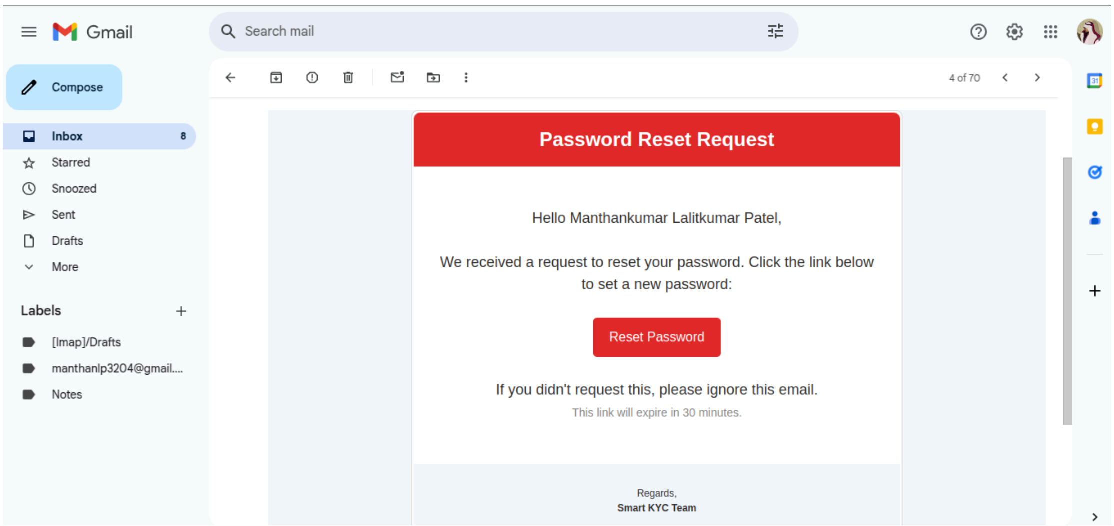

Fig 4.2.19 Password reset link email

5. Implementation

Smart KYC is a web-based KYC (Know Your Customer) verification system designed to provide a seamless, structured, and secure user experience. The platform simplifies the KYC process by enabling users to upload identity documents, extract relevant details using OCR (Optical Character Recognition), and perform facial verification to ensure authenticity. The system is developed using **FastAPI** for backend services and **React js** for a user-friendly frontend, ensuring efficiency, reliability, and ease of use.

- Landing Page
- Login Page
- Register Page
- Forgot Password Popup
- Verify OTP Popup
- Change Password Page
- Document Upload / KYC Verification Page
- Navbar
- Profile Management Section

5.1 Landing Page

The LandingPage is the welcome screen for Smart KYC. It includes a navbar, an introduction, a brief guide on the KYC process, and a "Get Started" button that navigates users to the document upload page.

5.2 Login Page

The LoginForm page handles user authentication for Smart KYC. It includes fields for email and password, a login button, and a "Forgot Password" option. The form sends credentials to the backend via an API request and stores the access token in cookies upon successful login. It also features a password visibility toggle, loading states, and navigation options for registration and password recovery.

5.3 Register Page

The RegisterForm page allows users to create an account for Smart KYC. It includes input fields for personal details such as name, email, phone number, date of birth, gender, and password. The form validates user input using Zod and submits registration data to the backend via an API call. It also features a password strength indicator, error handling, and navigation to the login page after successful registration.

5.4 Forgot Password Popup

The ForgotPasswordPopup is a modal component in Smart KYC that allows users to request a password reset. It includes an email input field and a button to send a reset link via an API request. The popup provides feedback using toast notifications and features a loading state for better user experience.

5.5 Verify OTP Popup

The VerifyOTP Popup is a modal component in Smart KYC for OTP verification. It includes an OTP input field, a "Verify OTP" button, and a "Resend OTP" option. The component handles OTP validation, API calls for verification and resending, and provides user feedback via toast notifications. It also features a loading state for better user experience.

5.6 Change Password Page

The ChangePassword page allows users to update their account password in Smart KYC. It includes fields for the current password, new password, and password confirmation. The form validates inputs using Zod, provides real-time password strength feedback, and sends the updated password to the backend via an API request. It also features error handling, a password visibility toggle, and redirects users to the login page after a successful password change.

5.7 Document Upload / KYC Verification Page

The DocumentUpload Page is used for uploading KYC documents in Smart KYC. It guides users through a multi-step process to upload Aadhaar (front & back), PAN card, and a selfie. The page includes:

- A progress indicator to show the current step.
- The UploadStep component for handling file selection and upload.
- API calls to process and verify uploaded documents.
- A Next/Finish button to proceed through steps or complete KYC verification.
- Loading state for better user experience.

5.8 Navbar

The Navbar component provides navigation for Smart KYC. It includes a mobile-friendly menu, site title and navigation links. If a user is authenticated, it displays a profile dropdown with options for profile, settings, changing passwords, and logging out. It also features a notification icon and login/register buttons for unauthenticated users.

5.9 Profile Management Section

The Profile Management page allows users to view and update their personal information, including profile photo, contact details, address, and KYC verification status. Users can:

- **View and Edit Personal Details**: Update full name, date of birth, gender, and address.
- **Profile Photo Management**: Upload a new profile picture or remove an existing one.
- **Email & Phone Verification**: Verify email and phone number with OTP-based authentication.
- **KYC Status Overview**: Check KYC verification status with visual indicators.
- **Address Management**: Edit and update address details, including city, state, country, and postal code.
- **Seamless User Experience**: A user-friendly interface with read-only and editable states for better control over profile updates.

6. Testing

System testing is a crucial aspect of Software Quality Assurance (SQA). It serves as the final validation phase to ensure that the Smart KYC system meets its functional, performance, and security requirements. Testing involves executing the system under various conditions to identify and fix errors before deployment. A well-structured test case is designed to uncover potential bugs and improve the system's reliability. The primary goal of testing is to verify that OCR-based document extraction, face-matching algorithms and user authentication work seamlessly and securely.

Tests:

6.1. User Registration Testing

Test ID	Input	Output	Expected Output	Test Verdict
TC_001	Valid registration data (Full Name, valid Email, valid Phone, DOB, valid Gender ID, Password)	User registration successful with a user ID	User registration successful with a user ID	Pass
TC_002	Registration data with duplicate email/phone	Email or phone number already exists.	Email or phone number already exists.	Pass
TC_003	Registration data with password less than 8 characters (e.g. "abc123")	At least 8 characters.	At least 8 characters.	Pass
TC_004	Registration data with invalid email (e.g. "user@invalid")	Invalid email address.	Invalid email address.	Pass

Table 6.1 User Register Testing Table

6.2. User Login Testing

Test ID	Input	Output	Expected Output	Test Verdict
TC_005	Valid credentials (registered email/phone and correct password)	Login successful. with access token	Login successful. with access token	Pass
TC_006	Non-existent email/phone	Invalid credentials.	Invalid credentials.	Pass
TC_007	Valid email with incorrect password	Invalid credentials.	Invalid credentials.	Pass

Table 6.2 User Login Testing Table

6.3. OTP Generation & Verification Testing

Test ID	Input	Output	Expected Output	Test Verdict
TC_008	Valid user email requesting OTP generation	OTP generates and sends successfully with OTP code.	OTP generates and sends successfully with OTP code.	Pass
TC_009	Incorrect OTP entered	Invalid OTP.	Invalid OTP.	Pass
TC_010	OTP expired or missing (OTP not found in Redis)	OTP is not found or expired.	OTP is not found or expired.	Pass
TC_011	Valid OTP provided that matches stored OTP	OTP verifies successfully.	OTP verifies successfully.	Pass

Table 6.3 OTP Generation & Verification Testing Table

6.4. Password Management Testing

Test ID	Input	Output	Expected Output	Test Verdict
TC_012	Valid user email for forgot password	The system sends the password reset email successfully.	The system sends the password reset email successfully.	Pass
TC_013	Valid reset token with new password provided	Password reset successful.	Password reset successful.	Pass
TC_014	Invalid/expired reset token with new password	Error from token verification (e.g., "Invalid token.")	Error from token verification (e.g., "Invalid token.")	Pass
TC_015	Valid current password and new password for change password	User changes the password successfully.	User changes the password successfully.	Pass
TC_016	Incorrect current password provided for change password	Current password is incorrect.	Current password is incorrect.	Pass

Table 6.4 Password Management Testing Table

6.5. Document Upload & Processing Testing

Test ID	Input	Output	Expected Output	Test Verdict
TC_017	Upload Aadhaar Front (JPG), valid user	Aadhaar front is being uploaded successfully with extracted data.	Aadhaar front is being uploaded successfully with extracted data.	Pass
TC_018	Upload Aadhaar Back (JPG), valid user	Aadhaar back is being uploaded successfully	Aadhaar back is being uploaded successfully with extracted data.	Pass

		with extracted data.		
TC_019	Upload PAN (JPG), valid user	Pan card is being uploaded successfully with extracted data.	Pan card is being uploaded successfully with extracted data.	Pass
TC_020	Upload Selfie (JPG), Aadhaar/PAN already uploaded	Face matches successfully, KYC is verified.	Face matches successfully, KYC is verified.	Pass
TC_021	Upload Selfie (JPG), Aadhaar/PAN uploaded but face mismatch	Selfie uploads successfully, but KYC is rejected due to mismatch.	Selfie uploads successfully, but KYC is rejected due to mismatch.	Pass
TC_022	Manually verify Aadhaar details after OCR	Document verifies successfully	Document verifies successfully	Pass

Table 6.5 Document Upload & Processing Testing Table

6.6. Edit Profile Testing

Test ID	Input	Output	Expected Output	Test Verdict
TC_023	Valid edit data (e.g., updated full name, phone, address details) for an existing user	User record updates with new details	User record updates with new details	Pass
TC_024	Edit data including "kyc_status_id" field	KYC status is not changeable	KYC status is not changeable	Pass
TC_025	Edit data with a new email that already exists in another user record	Email is already in use.	Email is already in use.	Pass
TC_026	Edit data with a new phone number that already exists in another user record	Phone number is already in use.	Phone number is already in use.	Pass

Table 6.6 Edit Profile Testing Table

6.7. Edit Profile Photo Testing

Test ID	Input	Output	Expected Output	Test Verdict
TC_027	Valid JPG/PNG, user with no photo	Profile photo uploads/updates successfully.	Profile photo uploads/updates successfully.	Pass
TC_028	Valid image, user with existing photo	Old photo replace with new photo	Old photo replace with new photo	Pass
TC_029	Invalid file (BMP/PDF), valid user	Invalid file format...	Invalid file format...	Pass
TC_030	Delete request, user with photo	Profile photo delete successfully.	Profile photo delete successfully.	Pass

Table 6.7 Edit Profile Photo Testing Table

7. Conclusion and Future Extensions

7.1 Recap and Achievements

The Smart KYC Verification System was conceptualized, designed, and developed to streamline the identity verification process using AI-powered tools. The goal was to build a secure and user-friendly platform that simplifies digital KYC for both users and businesses.

- **Conceptualization**: The idea was formed around reducing manual effort in KYC by using AI-based document scanning and facial recognition for efficient onboarding.
- **Design Phase**: System architecture and frontend-backend interaction flows were designed with scalability and modularity in mind.
- **Development**: The backend was developed using **FastAPI**, while the frontend was built using **React** to provide a modern and interactive user experience.

The system includes:

- OCR extraction from Aadhaar and PAN cards using OCR's & OpenAI's API
- Face matching using DeepFace to compare a user's selfie with their uploaded documents
- Dynamic document upload system with per-user file handling
- KYC status tracking and profile photo management
- User profile update features with validation and OTP-based verification of email.

Testing: Each major feature was accompanied by detailed test cases covering success and failure scenarios to ensure functionality, accuracy, and stability.

7.2 Future Enhancements

The current version lays a strong foundation for AI-powered KYC verification. Moving forward, the system can be enhanced with the following features:

WhatsApp-Based KYC Flow *(Planned)*: Integration with Twilio's WhatsApp API to allow users to complete their entire KYC process through chat, including uploading documents, receiving status updates, and verifying identity in a conversational interface.

Advanced Admin Dashboard: A React-based analytics dashboard to monitor document uploads, verification results, KYC statuses, and user activity.

Real-Time Government Verification: Integration with official APIs (e.g., UIDAI, NSDL) for live Aadhaar/PAN validation and fraud detection.

Multi-language OCR Support: Extend OCR capabilities to support regional languages to reach a broader audience.

7.3 Conclusion

The **Smart KYC Verification System** marks a significant step towards automating digital identity verification with the help of OCR & AI. It reduces the need for manual checks, speeds up onboarding, and creates a seamless experience for users. With a robust backend, a modern React frontend, and clear modularity for future integrations, the project is both functional and scalable.

While the current implementation covers core KYC workflows, planned features like WhatsApp-based interaction and real-time API verification will further strengthen the platform. We're proud of the journey so far and excited about the potential this system holds for the future of digital identity verification.

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